



workforce planning when AI changes the work

how to model demand and supply when task composition shifts, with a build, buy, borrow, bot matrix

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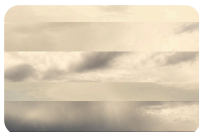
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the argument in short

Headcount planning assumes the job is the unit of demand. You forecast revenue, apply a ratio, and land on a number of people. That worked while jobs were stable containers for a stable set of tasks. Generative AI does not operate on jobs. It operates on tasks, unevenly, inside jobs that keep their titles. So the plan is built on the wrong unit, and it fails quietly. Nobody sees a role disappear. They see cycle times drift, backlogs move, and a hiring plan that no longer matches the work.

The regional evidence matters more than most planning decks admit. The ILO and NASK-PIB global index finds 34 per cent of employment in high-income countries sits in one of four generative AI exposure gradients, against 11 per cent in low-income countries. The IMF puts exposure at about 60 per cent in advanced economies, 40 per cent in emerging market economies and 26 per cent in low-income countries. That gradient runs straight through the markets covered here. Singapore and the Gulf states are high-income economies. India is lower-middle-income. Malaysia, Thailand, Indonesia, Vietnam and the Philippines sit in the upper-middle band. One global exposure figure applied across a regional workforce will be wrong in different directions in different countries.

The most useful finding here is arithmetic, not opinion. In a worked case of a 240-person finance shared services centre, two years of attrition at regional rates removes roughly 71 full-time equivalents of supply. The fastest task-absorption scenario removes about 63 full-time equivalents of demand over the same period. Supply falls faster than demand. The binding constraint in most functions is not redundancy. It is replacement, and specifically replacement of the capability the redesigned work now needs.

Four working instruments follow. The five-state task ledger converts a role inventory into defensible demand hours. The capability bridge separates the headcount gap from the capability gap, which usually differ by a factor of three or more. The four-door test scores build, buy, borrow and bot against six criteria and a three-year cost. The corner-four scenario sheet runs the plan under two drivers rather than one. Chapter six prices what happens when entry-level intake is cut. Chapter seven covers decision rights, a rerun trigger set and a sign-off pack. entomo publishes this as part of continuing work on how workforce planning is practised in Asian and Middle Eastern enterprises.

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01 CHAPTER

why headcount planning breaks when tasks move

The annual cycle forecasts roles. Generative AI redistributes tasks inside roles, so the plan misses the change entirely.

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Most large organisations plan headcount once a year, in a cycle owned jointly by finance and HR. Revenue and volume forecasts go in. A productivity ratio is applied, usually derived from last year. A number of people comes out, split by function and grade. The cycle is efficient because it treats the job as a fixed container. Twelve months later you check the variance and repeat.

That container assumption is now the weakest part of the model. Labour economists have described jobs as bundles of tasks for two decades. The ILO index built on that idea explicitly, scoring nearly 30,000 individual tasks rather than whole occupations. Exposure sits at the task level and varies sharply within a single job title. Two people with the same title, in the same team, can have very different task mixes.

the unit mismatch

When a technology changes tasks rather than jobs, three things happen that a headcount model cannot see. Time released inside a role does not automatically become a reduced headcount. Task composition shifts, so the role needs a different capability at the same grade. And output demand often rises, because cheaper output invites more of it. A model that forecasts only the number of people captures none of these.

3.3%

share of global employment in the ILO highest exposure gradient, where most tasks in the occupation carry high automation potential

ILO and NASK-PIB Working Paper 140, 2025

what the exposure numbers do and do not say

A much larger 20.5 per cent of global employment sits in the three lower gradients, where some tasks are exposed and others are not. Those are the jobs where planning gets difficult, because nothing disappears and everything changes. Two large independent studies give exposure by country income group. They use different methods and agree on the shape. Exposure rises with national income, reflecting the occupational mix of an economy rather than talent quality.

GENERATIVE AI EXPOSURE OF EMPLOYMENT BY COUNTRY INCOME GROUP, TWO INDEPENDENT ESTIMATES

Country group	ILO and NASK-PIB, 2025 (share in any exposure gradient)	IMF, 2024 (share of employment exposed)
High income / advanced economies	34%	About 60%
Emerging market economies	Not reported separately	40%
Low income	11%	26%
World	About 24 to 25%	Almost 40%

Apply this to a regional footprint. A shared services centre in Bengaluru, a regional headquarters in Singapore and a corporate function in Dubai are not on the same exposure curve. The World Bank classifies India as lower-middle-income. Singapore, the United Arab Emirates and Saudi Arabia are high-income. In the 2026 to 2027 classification the Philippines and Vietnam moved up into the upper-middle band, joining Malaysia, Thailand and Indonesia.

READING NORTH AMERICAN STUDIES IN ASIAN AND GULF MARKETS

The best labour-market evidence on AI effects is currently American, because American payroll data is available at high frequency. The Stanford Digital Economy Lab work uses ADP records covering millions of United States workers. Its findings describe a high-income labour market with high AI adoption, thin employment protection and no nationalisation quotas. None of those conditions hold uniformly across Southeast Asia, India or the Gulf. Use that evidence for mechanism, which is how the effect travels. Do not use it for magnitude.

The mechanism is portable and worth taking seriously. Employment among workers aged 22 to 25 in highly AI-exposed occupations sits about 19 per cent below where it would be had it tracked their less-exposed peers. The adjustment runs primarily through reduced hiring rather than increased separations. Experienced workers show no comparable gap. That is a hiring-channel effect, and hiring channels are what a workforce plan controls.

three ways the annual cycle fails

- **It nets out.** A function loses 40 full-time equivalents of routine demand and gains 15 of new oversight, review and exception demand. The headcount line shows minus 25 and the capability change is invisible.
- **It uses last year's ratio.** The productivity ratio is derived from a period in which the task mix was different. Applying it forward carries the old task mix into the new plan.

- **It has one scenario.** A single-point forecast forces a single decision. When the assumption moves, the plan is not adjusted, it is abandoned, and hiring reverts to manager requests.

29%

share of chief human resources officers confident they can deliver on strategic workforce planning goals

Gartner, cited in Deloitte Insights, 2025



A plan that forecasts only the number of people will be right about the total and wrong about every part of it.

The rest of this document is a method. It starts with demand at the task level, because every other number in a workforce plan depends on getting that one right.



02 CHAPTER

modelling demand at the task level

Convert roles into task hours, assign each task a state with an evidenced multiplier, then add back the demand that growth creates.

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Task-level demand modelling has a reputation for being impossibly detailed. It is not, provided you resist the temptation to model every task. You need roughly six to nine task groups per role family, each carrying at least five per cent of the hours. Anything smaller is noise. A function of 250 people can be modelled properly in about three weeks of analyst time.

build a task inventory you can defend

1. Start from an external taxonomy rather than internal job descriptions. The O*NET database publishes task statements by occupation, free, and gives you a neutral starting vocabulary that HR did not write.
2. Reconstruct time allocation from systems, not surveys. Ticket volumes, case counts, transaction logs and calendar data give you hours per task group with far less bias than asking people to estimate.
3. Reconcile to total paid hours. If the task groups sum to 130 per cent of available hours, the inventory is wrong and everything downstream inherits the error.
4. Have the line manager challenge the split before you model anything. The purpose is not agreement. It is to surface the tasks that systems do not capture, such as escalation, coaching and rework.

Use a productive hours figure, not a contractual one. A common convention is 1,700 productive hours per full-time equivalent per year, after leave, holidays, training and administrative time. Whatever number you choose, state it, apply it consistently, and do not change it between scenarios.

the five-state task ledger

Every task group is assigned exactly one of five states, each carrying a demand multiplier applied to its current hours.

01 **steady, multiplier 1.00**

The task is unchanged in method and volume per unit of output. Judgement-heavy, relationship-heavy and physically present work usually lands here. Most functions are 25 to 40 per cent steady, and planners consistently underestimate this share.

02 **assisted, multiplier 0.90**

A person still performs the task end to end, with drafting or retrieval support. The gain is real but modest, typically 5 to 15 per cent, and it is spent partly on checking. Assume 0.90 unless you have measured otherwise.

03 **compressed, multiplier 0.70**

The task is restructured so the person reviews and corrects rather than produces. This is where most of the genuine gain sits. It requires a workflow change, not a tool licence, which is why it takes longer to realise than the business case assumes.

04 **rerouted, multiplier 0.50**

The task leaves the team. It moves to self-service, to the requester, or to a different function. The hours do not vanish from the enterprise. Record where they land, or you will book the same saving twice.

05 **retired, multiplier 0.15**

The task stops being performed by this team in recognisable form. Reserve a residual of 0.15 for oversight, exception handling and the cases the system refuses. A multiplier of zero is almost always wrong.

The multipliers are the contested part, so they must be defensible. The published evidence on generative AI productivity effects is wider than most business cases acknowledge, and the spread is itself informative. Effects depend heavily on task type and on the prior skill of the worker.

MEASURED PRODUCTIVITY EFFECTS, FOUR INDEPENDENT STUDIES

Study	Setting	Measured effect
Generative AI at Work, QJE, 2025	5,179 customer support agents, staggered rollout	About 14% more issues resolved per hour on average; 34% for novice and low-skilled agents; minimal effect for experienced agents
METR randomised controlled trial, 2025	16 experienced open-source developers, 246 tasks in repositories they knew well	Tasks took 19% longer with AI tools allowed; the same developers estimated afterwards that AI had made them 20% faster
GPTs are GPTs, Science, 2024	Task-level exposure rubric applied to O*NET	About 80% of the US workforce could have at least 10% of tasks affected; about 19% could see at least 50% affected
Anthropic Economic Index, 2026	Classified usage patterns on a large assistant platform	Augmentation patterns 52% of conversations against automation patterns 45%, a reversal of the August 2025 position

Three lessons follow. Gains concentrate among less experienced workers, which has direct consequences for the entry-level question in chapter six. Gains are not guaranteed, and can be negative for expert work in familiar contexts. And the dominant observed pattern remains augmentation rather than substitution, which argues for the assisted and compressed states over retired.

a worked example

Take a finance shared services centre of 240 full-time equivalents. At 1,700 productive hours each, annual capacity is 408,000 hours. The task inventory produced seven groups. Each is assigned one state, with the multiplier applied to current hours.

FIVE-STATE TASK LEDGER, FINANCE SHARED SERVICES CENTRE, 240 FULL-TIME EQUIVALENTS

Task group	Share of hours	Current hours	State and multiplier	Modelled hours
Invoice matching and exception handling	28%	114,240	compressed 0.70	79,968
Vendor and customer queries	18%	73,440	compressed 0.70	51,408
Month-end close and reconciliations	16%	65,280	assisted 0.90	58,752
Reporting and commentary	12%	48,960	retired 0.15	7,344
Master data maintenance	10%	40,800	rerouted 0.50	20,400
Audit support and controls testing	9%	36,720	steady 1.00	36,720
Leadership, training and escalation	7%	28,560	steady 1.00	28,560
Total	100%	408,000		283,152

The arithmetic is straightforward. Modelled hours of 283,152 divided by 1,700 gives 166.6 full-time equivalents. Against 240 today, that is a reduction of 73.4, or 30.6 per cent. This is the number that gets into a slide, and on its own it is misleading.

THE VOLUME TRAP

Cheaper output raises demand for output. If a reconciliation now costs a quarter of what it cost, the business will ask for more of them, sooner, in more cuts. Task-level demand modelling that ignores volume elasticity systematically overstates reduction. Apply growth to volume-driven task groups only. Leadership, audit and controls hours do not scale with transaction count in the same way.

Continue the example over two years, with transaction volume growing 9 per cent a year. Of the 283,152 modelled hours, 210,528 sit in volume-driven groups, which is 74.3 per cent. Two years of compounding gives a factor of 1.09 squared, or 1.1881. So 210,528 multiplied by 1.1881 is 250,128 hours. Adding back the 72,624 non-volume hours gives 322,752 hours, or 189.9 full-time equivalents.

One adjustment remains. The redesign creates new task hours that did not exist: model output review, exception adjudication, workflow maintenance and control testing on automated steps. Estimate this at 6 full-time equivalents, which is 10,200 hours. We are estimating here, because there is no benchmark we would defend. It is roughly 3.5 per cent of modelled hours, and we would sanity check it against the oversight ratio the risk function will accept. Total demand becomes 332,952 hours, or 195.9 full-time equivalents.

So the honest answer is 240 today to about 196 in two years. That is a 44 full-time equivalent reduction, or 18.3 per cent, against a first-pass figure of 30.6 per cent. The difference of 12 percentage points is entirely volume growth and new task creation. Both were knowable at the start.



03 CHAPTER

modelling supply: attrition, internal movement and skill decay

Supply leaks in three places and is refilled in two, and the headcount gap is almost never the same as the capability gap.

— EBOOK · ENTOMO RESEARCH

Supply modelling in most organisations is a single line: opening headcount minus expected attrition. That line hides more than it shows. People leave the enterprise, they leave the function for another function, and they stay in the role while the role changes underneath them. Only the first is usually modelled.

FRAMEWORK

the capability bridge

Walk opening headcount to closing capable supply through six steps, and report both the headcount gap and the capability gap.

01 1. opening headcount

Actual heads in role at period start, not budgeted or approved positions. Use the payroll figure, because approved positions overstate supply by whatever the vacancy rate happens to be.

02 2. external exits

Apply an attrition rate specific to this country, function and grade band. A single enterprise-wide rate will be wrong everywhere. Split voluntary from involuntary, because only voluntary responds to retention spend.

03 3. internal transfers out

People moving to other functions. This is real supply loss for you and real supply gain for someone else. If it is not modelled, the enterprise double counts the same person in two plans.

04 4. internal transfers in

Inbound movement from other functions, including redeployment from areas the plan is shrinking. This is the leg the build, buy, borrow, bot matrix in chapter four exists to feed.

05 5. capability assessment

Of the people you will still have, how many meet the capability standard the redesigned task set requires. This is the step that turns a headcount plan into a workforce plan.

06 6. development throughput

How many of the remainder can be brought to standard within the horizon, given actual learning capacity and actual manager time. Constrain this to what the function has previously delivered, not to what the learning catalogue offers.

attrition is not one number

The regional benchmarks are public, current and specific enough to plan with. They also show why an enterprise-wide assumption fails. A 16 per cent India assumption applied to a Manila operation understates exits by four percentage points, which on 800 people is 32 people a year. Note also that falling attrition is not good news for a plan that assumes reduction, because fewer voluntary exits means less natural capacity release.

Market	Attrition rate	Source and basis
India, all industries	16.2% in 2025, from 17.7% in 2024 and 18.7% in 2023	Aon Annual Salary Increase and Turnover Survey, more than 1,400 organisations across 45 industries
Philippines	20.0% projected	Aon Southeast Asia study, more than 700 businesses, fieldwork July to September 2025
Singapore	19.3% projected	Aon Southeast Asia study, 2025
Malaysia	18.2% projected	Aon Southeast Asia study, 2025
Consulting, business and community services, Southeast Asia	22.6%, the highest industry rate in the region	Aon Southeast Asia study, 2025

16.2%

India attrition in 2025, the lowest in the series and down 2.5 points in two years

Aon Annual Salary Increase and Turnover Survey, 2026

internal movement is a policy variable, not a fact

Internal hiring is the cheapest supply channel and the most neglected. Analysis of almost 600,000 hire-related data points found internal hires accounted for about 24 per cent of all hires, down from around 40 per cent at the 2020 peak and below the 30 to 32 per cent historical baseline. Internal moves also compress time to hire by roughly 10 to 12 days. Treat the internal mobility rate as a lever you set. If your plan requires 60 cross-function moves and your organisation has never exceeded 25 in a year, that is a delivery problem no demand model will fix.

skill decay, handled honestly

You will be offered a figure for the half-life of skills, usually five years, or two and a half for technical skills. We do not use it and recommend you do not either. The figure circulates widely without a traceable measurement behind it. Building a supply model on an unmeasured constant is worse than building one on an assumption you have labelled as an assumption. What is measurable is the rate at which required skills change. LinkedIn reports that 70 per cent of the skills used in most jobs will change by 2030, and that since 2022 the rate at which members add new skills has risen 140 per cent. The World Economic Forum survey of more than 1,000 employers puts it at 39 per cent of core skills changing by 2030. These measure requirement drift, which is what your model needs.

39%

share of core skills employers expect to change by 2030, across 1,000 employers representing more than 14 million workers

World Economic Forum Future of Jobs Report, 2025

Convert requirement drift into a testable annual assumption. If 39 per cent of core skills change over five years, a flat annualised approximation is 7.8 per cent a year, since 39 divided by 5 is 7.8. Apply that as annual erosion of assessed capability, hold it constant across scenarios, then check it against your own reassessment data after twelve months. The point is not that 7.8 per cent is correct. The point is that it is written down, applied consistently and falsifiable.

the bridge, worked, for a 1,000-person function in India

CAPABILITY BRIDGE OVER A 12-MONTH HORIZON, TECHNOLOGY AND OPERATIONS FUNCTION OF 1,000 PEOPLE, AGAINST A MODELLED REQUIREMENT OF 880 CAPABLE FULL-TIME EQUIVALENTS

Step	Calculation	Result
Opening headcount	Payroll actual	1,000
External exits	1,000 multiplied by 16.2%	minus 162, giving 838
Internal transfers out	4% of opening headcount	minus 40, giving 798
Internal transfers in	Committed inbound moves	plus 25, giving 823
Capable against new task set	823 multiplied by 62% assessed capable	510
Development throughput	313 not yet capable, 45% brought to standard in 12 months	plus 141, giving 651
Demand	From the task ledger	880
Headcount gap	880 minus 823	57
Capability gap	880 minus 651	229

The headcount gap says 57. The capability gap says 229. They differ by a factor of four. A plan that reports only the first will hire 57 people and remain 172 capable full-time equivalents short at year

end. That is not a forecasting error. It is a category error, and it is the single most common defect we see in enterprise workforce plans.



**The headcount gap tells you how many chairs are empty.
The capability gap tells you how much of the work will
not get done.**



04 CHAPTER

the build, buy, borrow, bot decision matrix

Four doors close every capability gap, they trade cost against speed in opposite directions, and the choice should be scored rather than argued.

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Once the capability gap is quantified, there are only four ways to close it. Each has a characteristic cost profile, a characteristic time to capability and a characteristic failure mode. The purpose of a matrix is to stop the decision being made by whoever presents last.

- **Build.** Develop existing employees into the capability. Slowest to start, cheapest at steady state, strongest on retention and institutional knowledge. Fails when development capacity is asserted rather than evidenced.
- **Buy.** Hire the capability externally. Fast, expensive, and carries the market premium plus a ramp period. Fails when the capability is scarce in the local market or blocked by nationalisation quotas.
- **Borrow.** Contract, second or use a managed service. Fastest to capability, highest unit cost, no permanent liability. Fails on knowledge retention, because the capability leaves when the contract ends.
- **Bot.** Automate the task so the capability requirement falls. Highest fixed cost, lowest marginal cost, longest lead time. Fails more often than business cases assume.

FRAMEWORK

the four-door test

Score each door from 1 to 5 on six criteria, weight the criteria to the decision at hand, and record the score alongside the eventual outcome.

01 **criterion 1, time to capability**

Weeks until the capability is producing at standard, not until the contract is signed or the course is booked. Score 5 for under eight weeks, 1 for over twelve months.

02 **criterion 2, three-year fully loaded cost**

All-in cost over the planning horizon, including ramp, productivity dip, recruitment fees, rework allowance and management overhead. Score 5 for the cheapest door, 1 for the most expensive.

03 **criterion 3, durability**

Whether the capability remains with the enterprise at the end of the horizon. Build and bot score high, borrow scores low. This criterion is where most short-horizon decisions destroy value.

04 **criterion 4, reversibility**

Cost of exit if the assumption is wrong. Borrow is highly reversible, bot is not. Weight this heavily when the demand forecast has wide error bars.

05 **criterion 5, regulatory and quota fit**

Whether the door is available under local employment rules. In several markets it is not. Score 1 where the door creates a compliance exposure regardless of its other merits.

06 **criterion 6, delivery confidence**

The organisation's own track record on this door. If the last three automation programmes slipped two quarters, that is the evidence, and it belongs in the score.

a worked cost comparison

Requirement: 20 full-time equivalents of a redesigned claims adjudication capability, three-year horizon. Fully loaded cost of the relevant grade is taken as USD 22,000 per year. All figures below are illustrative and built from stated assumptions, so substitute your own rates.

THREE-YEAR FULLY LOADED COST, FOUR DOORS, 20 FULL-TIME EQUIVALENTS OF CAPABILITY

Door	Cost build-up	Three-year total	Time to capability
Build	Base pay 20 by 22,000 by 3 = 1,320,000. Training 20 by 3,000 = 60,000. Productivity dip 15% for six months = 20 by 0.15 by 0.5 by 22,000 = 33,000. Requalification uplift 8% in years 2 and 3 = 20 by 22,000 by 0.08 by 2 = 70,400.	USD 1,483,400	6 to 9 months
Buy	Base pay at an 18% market premium = 20 by 22,000 by 1.18 by 3 = 1,557,600. Recruitment at 15% of first-year salary = 77,880. Ramp, four months at 50% productivity = 86,533. Replacement of 30% of the cohort over three years = 49,326.	USD 1,771,339	3 to 5 months
Borrow	Contract rate at a 45% premium = 20 by 22,000 by 1.45 = 638,000 a year. Full volume in years 1 and 2, half in year 3 = 1,595,000. Vendor management overhead at 5% = 79,750.	USD 1,674,750	4 to 8 weeks
Bot	Platform and integration build 420,000. Run cost 90,000 a year by 3 = 270,000. Retained oversight of 6 full-time equivalents = 6 by 22,000 by 3 = 396,000. Change and adoption 110,000. Rework allowance at 30% of build = 126,000.	USD 1,322,000	9 to 14 months

The ranking on three-year cost is bot, build, borrow, buy. The ranking on time to capability is the exact reverse: borrow, buy, build, bot. That inversion is the whole point of the matrix. Any door can be justified by choosing which of the two rankings to present. A scored decision forces both onto the same page.

THE BOT DOOR IS NOT FREE AND FREQUENTLY DOES NOT OPEN

The 30 per cent rework allowance in the table is not conservatism. Survey evidence from more than 1,000 enterprises found that 42 per cent abandoned most of their AI initiatives in 2025, up from 17 per cent the year before, and that the average organisation scrapped 46 per cent of AI proofs of concept before they reached production. If your bot business case carries no failure allowance, it is not a business case. It is a hope with a spreadsheet.

46%

share of AI proofs of concept scrapped before reaching production, average across surveyed enterprises

S&P Global Market Intelligence, 2025

the regional overlay

Two of the four doors are constrained by national policy in these markets, and the constraint is a hard input rather than a risk. In the United Arab Emirates, private-sector companies with 50 or more employees must raise Emiratisation in skilled roles by 2 percentage points a year, 1 point by 30 June and 1 by 31 December, to reach 10 per cent. Shortfalls carry a monthly financial contribution per unfilled position, which has been rising annually.

Saudi Arabia operates profession-level Saudisation quotas under Nitaqat, with minimum salary thresholds for a national employee to count toward the quota. In practice this means the buy door in the Gulf is not a single door. It is two doors with different availability, different cost and different compliance consequences. A plan that models one blended external hire is not modelling the market it is in.

The borrow door has the opposite profile in India, where it is unusually deep. The government's own projection puts gig and platform workers at 23.5 million by 2029 to 2030, up from 7.7 million, moving from 1.5 per cent to 4.1 per cent of the workforce. Roughly 22 per cent of that work is classified as high-skilled and 47 per cent as medium-skilled. That is a genuine capacity pool for time-boxed capability, and it is materially larger than it was when most sourcing policies were written.

A working rule. Use borrow to buy time while build runs. Use buy only where the capability cannot be developed inside the horizon. Use bot only where the task is high volume, stable in specification and already instrumented. Use build for anything you expect to still need in three years.



05

CHAPTER

four scenarios and the template for running them

Two drivers produce four corners, and the corners disagree with each other in ways that a single forecast conceals.

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Scenario planning fails in most organisations for a boring reason. Teams build a base, a high and a low case, which are the same forecast at three confidence levels. That produces one story with error bars. It does not produce different futures, so it does not produce different decisions.

Use two independent drivers instead, and take all four corners seriously. For workforce planning under task change, the two drivers that matter are the speed at which tasks are absorbed, and the direction of demand for the function's output. They are genuinely independent. Fast absorption in a growing business is a different problem from fast absorption in a flat one.

FRAMEWORK

the corner-four scenario sheet

One page per scenario, four pages per function, with the same fields in the same order so the four can be read against each other.

01 field 1, the two driver settings

State the absorption factor and the volume growth rate as numbers, not adjectives. Slow and fast mean nothing in a review. 0.92 and 0.69 can be argued with.

02 field 2, demand in full-time equivalents

Output of the five-state task ledger under this scenario's absorption factor, with volume growth applied to volume-driven hours only.

03 field 3, supply in full-time equivalents

Output of the capability bridge under this scenario's attrition and mobility assumptions, with no hiring, so the natural position is visible.

04 field 4, the gap and its sign

Demand minus supply. Record the capability gap separately from the headcount gap, as in chapter three.

05 field 5, the door mix

How the gap is closed under the four-door test, expressed as full-time equivalents per door and cost per door.

06 field 6, the leading indicator

The one observable that tells you within a quarter whether this scenario is the one you are in. Without this field, the sheet is decoration.

07 field 7, the decision this scenario forces

The specific action that differs from the other three corners, and the date by which it must be taken to be useful.

Return to the 240 full-time equivalent finance shared services centre from chapter two. Absorption is set at 0.92 for slow and 0.69 for fast. Volume growth is 2 per cent a year for flat and 9 per cent for growing. Volume-driven hours remain 74 per cent of the total. New task creation is added per scenario. The horizon is two years.

CORNER-FOUR DEMAND, FINANCE SHARED SERVICES CENTRE, TWO-YEAR HORIZON

Scenario	Absorption factor	Volume growth	Demand calculation	Demand, full-time equivalents
Hold: slow absorption, flat volume	0.92	2% a year, factor 1.0404	240 by 0.92 = 220.8. Volume-driven 163.4 by 1.0404 = 170.0, plus 57.4 non-volume = 227.4, plus 3 new-task	230
Squeeze: fast absorption, flat volume	0.69	2% a year, factor 1.0404	240 by 0.69 = 165.6. Volume-driven 122.5 by 1.0404 = 127.5, plus 43.1 non-volume = 170.6, plus 6 new-task	177
Stretch: slow absorption, growing volume	0.92	9% a year, factor 1.1881	240 by 0.92 = 220.8. Volume-driven 163.4 by 1.1881 = 194.1, plus 57.4 non-volume = 251.5, plus 4 new-task	255
Reshape: fast absorption, growing volume	0.69	9% a year, factor 1.1881	240 by 0.69 = 165.6. Volume-driven 122.5 by 1.1881 = 145.6, plus 43.1 non-volume = 188.7, plus 8 new-task	197

Now the supply side, held constant across all four so the scenarios remain comparable. With attrition at 16 per cent a year and no hiring, supply after two years is 240 multiplied by 0.84 squared. That is 240 multiplied by 0.7056, or 169 full-time equivalents. The gap in each corner is demand minus 169.

Scenario	Demand	Supply with no hiring	Gap	What it forces
Hold	230	169	61 short	Sustained replacement hiring. The AI question is not yet the binding one.
Squeeze	177	169	8 short	Near balance. Redeploy rather than exit, and stop backfilling selectively.
Stretch	255	169	86 short	Aggressive multi-door sourcing. This is the most expensive corner, and it is the one least often modelled.
Reshape	197	169	28 short	Targeted hiring into new task capability while routine intake stops.

Read the four together and the central finding appears. Attrition over two years removes 71 full-time equivalents of supply, since 240 minus 169 is 71. The fastest absorption scenario removes 63 full-time equivalents of demand, since 240 minus 177 is 63. Supply falls faster than demand in every corner. There is no scenario in this case in which the function is oversupplied.



In three of four corners the organisation is short of people. In the fourth it is short by eight. A plan built to manage reduction would be wrong in all four.

NO-REGRET MOVES

The actions that appear in all four corners should be started immediately and taken out of the scenario debate. In this case: reskill the reporting and commentary population, because that task group is retired or heavily compressed in every corner; hold entry-level intake, because every corner needs replacement; and instrument the invoice exception queue, because the leading indicator for absorption speed lives there.

scenario sheet completeness check

- ☐ Both driver settings are expressed as numbers, and the numbers differ meaningfully between corners
- ☐ Demand and supply use the same productive hours convention and the same attrition source
- ☐ Volume growth is applied only to volume-driven task groups, and the volume-driven share is stated
- ☐ New task creation is added explicitly and labelled as an estimate where it is one
- ☐ The supply line shows the no-hiring position, so the natural trajectory is visible before intervention
- ☐ Every corner has one named leading indicator observable within a quarter
- ☐ Every corner names one decision that differs from the other three, with a date
- ☐ No-regret moves are listed separately and have owners, not just names



06 CHAPTER

the entry-level problem and the pipeline it breaks

Cutting junior intake balances this year's budget and creates a senior shortage five years out that nobody will attribute to the decision.

EBOOK · ENTOMO RESEARCH

Entry-level work is unusually exposed, for a structural reason. Junior roles concentrate the tasks that are most codified, most supervised and most easily specified. Those are precisely the tasks that generative AI performs best. So the exposure lands first on the population the organisation is least likely to defend in a budget conversation.

19%

gap in employment for United States workers aged 22 to 25 in highly AI-exposed occupations, against similarly aged peers in less-exposed occupations

Stanford Digital Economy Lab, 2026

The detail matters more than the headline. That gap widened from about 15 per cent in July 2025 to about 19 per cent by June 2026. In absolute terms, employment of 22 to 25 year olds in the two most exposed quintiles fell roughly 11 per cent between November 2022 and June 2026, while the same age group in less-exposed occupations grew about 10 per cent. Experienced workers show no comparable gap.

Two caveats are essential before this travels. The data is United States payroll data from a single large provider. The United States is a high-income labour market with high AI adoption, no nationalisation quotas and thin dismissal protection. The mechanism, which is adjustment through reduced hiring rather than increased separations, is likely to be general. The magnitude is not.

the evidence outside the United States is thinner and more mixed

United Kingdom data illustrates both the pattern and the difficulty of attribution. Online job advertisements fell 31 per cent from the three months ending May 2022, with a 38 per cent reduction in roles heavily exposed to AI against 21 per cent in low-exposure sectors. Graduate job postings dipped by around 8 per cent year on year, but apprentice hiring rose 8 per cent, so the net decline across entry-level pathways was about 5 per cent. That last figure is the honest one, and it is much smaller than the headline. It also arrives alongside a labour market correction after pandemic-era over-hiring, rising employment costs and weak employer confidence. Attributing the whole movement to AI is not supported by the data. Attributing none of it is equally unsupported.

WHAT DIFFERENT ENTRY-LEVEL SOURCES REALLY MEASURE

Source and year	What it measures	What it cannot tell you
Stanford Digital Economy Lab, 2026	Actual employment of 22 to 25 year olds by occupation exposure, from United States payroll records	Whether the same magnitude holds in markets with different adoption, cost structures and labour law
McKinsey, cited by techUK, 2025	Change in volume of online job advertisements by AI exposure, United Kingdom	Whether advertisements translate into hires, or whether hiring simply moved to other channels
Institute of Student Employers, cited by techUK, 2025	Employer-reported graduate and apprentice hiring, United Kingdom	The cause of the change, which includes cost, confidence and post-pandemic correction
NASSCOM Strategic Review, 2026	India technology sector direct employment, at roughly 5.95 million with net addition of about 135,000 in FY26, up about 2.3 per cent	Composition by grade, so entry-level share must be inferred rather than read

India shows a different shape again. Technology sector employment is still growing, at roughly 2.3 per cent, while revenue growth is materially faster. That divergence is the signal. Headcount elasticity of revenue is falling, which is a productivity story rather than a redundancy story. Adoption is high: 92 per cent of Indian knowledge workers reported using AI at work against 75 per cent globally in the 2024 Work Trend Index.

why this is a planning problem and not an HR problem

Senior capability is grown, not bought, in most functions. External senior hires carry a market premium, a longer ramp and a higher first-year exit risk. The internal pipeline exists to avoid paying that premium. When junior intake is cut, the pipeline does not fail immediately. It fails at the point where today's juniors would have become tomorrow's seniors, which is typically four to six years later.

FRAMEWORK

the bench replacement ratio

One number that tells you what share of senior replacement need your own pipeline can cover, and therefore what share you will have to buy.

01 **step 1, count senior replacement need**

Senior population multiplied by senior attrition, plus the annual increase in senior demand. This is how many senior people you must find each year.

02 **step 2, count junior intake**

Annual intake into the entry grade that feeds this senior population, whether graduate, apprentice or trainee.

03 **step 3, apply survival to seniority**

Junior retention rate raised to the power of years to seniority. Then multiply by the share of survivors who actually qualify for the senior grade.

04 **step 4, divide**

Pipeline supply divided by senior replacement need. A ratio of 1.0 means self-sufficiency. Below 0.5 means the majority of your senior population will be bought at market rates.

05 **step 5, run it twice**

Once at current intake and once at proposed intake. The difference, expressed in additional external senior hires a year, is the true cost of the intake decision.

Worked example. A 1,200-person engineering organisation with a senior population of 400. Senior attrition is 14 per cent, giving 56 exits a year. Senior demand grows 2 per cent, adding 8. Annual senior replacement need is therefore 64.

BENCH REPLACEMENT RATIO AT TWO INTAKE LEVELS, FIVE YEARS TO SENIORITY

Line	At current intake of 90	At proposed intake of 45
Junior intake a year	90	45
Junior annual retention	78%	78%
Survival over five years, 0.78 to the power of 5	0.2887	0.2887
Survivors reaching year five	90 by 0.2887 = 26.0	45 by 0.2887 = 13.0
Share of survivors qualifying for senior grade	80%	80%
Pipeline supply of seniors a year	26.0 by 0.8 = 20.8	13.0 by 0.8 = 10.4
Senior replacement need a year	64	64
Bench replacement ratio	20.8 divided by 64 = 0.33	10.4 divided by 64 = 0.16
External senior hires required a year	64 minus 20.8 = 43	64 minus 10.4 = 54

Halving intake adds about 10 external senior hires a year, from 43 to 54. If a senior external hire costs 25 per cent more in loaded terms than an internal promotion, and a senior costs USD 45,000 loaded, the premium is USD 11,250 per hire. Ten additional hires is USD 112,500 a year, recurring, starting in year five. The intake saving that produced it appears in year one. No accounting system will connect the two.

THE AUGMENTATION PARADOX

The strongest measured productivity gains from generative AI accrue to novice and low-skilled workers. In the customer support study, novice and low-skilled agents gained 34 per cent against an average of about 14 per cent, with minimal gains for experienced agents. Junior workers are simultaneously the group most exposed to task displacement and the group that benefits most from the technology. That is not a contradiction. It is an argument for redesigning entry-level roles rather than deleting them.

The design implication is specific. Keep intake, change what the intake does. Move junior roles up the task ledger from production to review, exception handling and quality assurance. Those tasks build the judgement that seniority requires, and they are the tasks the five-state ledger shows growing rather than shrinking.



07 CHAPTER

governance: who signs the plan and when it is rerun

Workforce plans rarely fail on arithmetic, they fail because nobody owns an assumption and nothing forces a rerun.

EBOOK · ENTOMO RESEARCH

A plan with no named owner per assumption is a document, not a plan. The governance question is not who approves the total. It is who is accountable for each input, who is entitled to challenge it, and what event obliges the group to reconvene before the next annual cycle.

who signs what

Separate four rights explicitly: the right to propose an assumption, the right to challenge it, the right to sign it, and the right to be told. Confusing propose with sign is the most common governance failure, and it usually appears as finance signing task assumptions it has no basis to evaluate.

DECISION RIGHTS FOR A TASK-LEVEL WORKFORCE PLAN

Decision	Proposes	Challenges	Signs	Informed
Task inventory and time allocation	Function line leadership	People analytics	Function head	Finance
Task state assignment and multipliers	People analytics with function leads	Finance and technology	Function head	Risk and audit
Volume growth and elasticity assumptions	Finance	Function line leadership	Finance business partner	People analytics
Attrition, mobility and development throughput	People analytics	Finance	Chief human resources officer	Function head
Door mix under the four-door test	Workforce planning lead	Finance and procurement	Joint, function head and finance	Technology and legal
Quota and compliance constraints by market	Country HR lead	Legal	Country HR lead	Chief human resources officer and finance
Budget release against the plan	Finance	Function head	Finance	Chief human resources officer
Call to rerun the plan	Workforce planning lead	Any signing party	Chief human resources officer and chief financial officer jointly	Executive committee

The last row is the one that gets deleted from most governance models, and it is the one that determines whether the plan survives contact with the year. If nobody can call a rerun, the plan is annual by default, and an annual plan cannot track a change that moves quarterly.

the rerun trigger set

Eight observable conditions, each with a numeric threshold, that oblige the signing group to reconvene within 15 working days.

01 1. volume trigger

Forecast volume for any modelled driver moves more than 10 per cent from the planning assumption on a rolling three-month basis.

02 2. absorption trigger

Any task group moves state earlier than planned, or a deployment reaches more than 60 per cent of its target user base ahead of schedule.

03 3. attrition trigger

Rolling twelve-month attrition in any country or function moves more than 3 percentage points from the planning assumption.

04 4. pipeline trigger

Entry-level intake or internal promotion rate falls more than 20 per cent below plan. This trigger exists because pipeline damage is invisible for years without it.

05 5. regulatory trigger

A nationalisation quota, visa rule, minimum salary threshold or labour law changes in any market where the plan assumed a sourcing channel.

06 6. delivery trigger

A build or bot programme misses a capability milestone by more than one quarter. Given that a large share of AI proofs of concept never reach production, this trigger will fire.

07 7. cost trigger

Fully loaded cost per capable full-time equivalent moves more than 8 per cent against plan in any door.

08 8. accuracy trigger

Forecast error on any function in the last completed period exceeds 10 per cent. Persistent error means the model is wrong, not that reality misbehaved.

cadence

1. Monthly, at analyst level: refresh the eight trigger measures and publish them unchanged, whether or not any has fired. Suppressing a fired trigger for one month is the most common way plans quietly die.
2. Quarterly, at signing group level: review triggers, reassess capability against the current task set, and confirm or change the door mix. No new demand model unless a trigger has fired.

3. Half-yearly: rerun the corner-four scenario sheet and check which corner the leading indicators say you are in. Retire corners that are no longer plausible and add one if a new driver has appeared.
4. Annually: rebuild the task inventory from source, not from last year's file. Task inventories drift, and a second-generation copy of a copy is worse than no inventory.

the sign-off pack

- ☐ Task inventory with source of time allocation stated per task group, and the productive hours convention used
- ☐ State assignment per task group with the multiplier and the evidence or assumption behind it, marked as measured or estimated
- ☐ Demand in full-time equivalents, showing task change and volume growth as separate lines rather than one net figure
- ☐ Supply as a capability bridge, with headcount gap and capability gap reported separately
- ☐ Attrition, mobility and development throughput assumptions with their source, date and geography
- ☐ Four-door scores and three-year costs for every gap above a stated materiality threshold
- ☐ Four scenario sheets with named leading indicators and the decision each corner forces
- ☐ Quota and compliance constraints per market, signed by the country HR lead
- ☐ Last period's forecast error by function, with an explanation of any error above 10 per cent
- ☐ The trigger dashboard, showing current value against threshold for all eight triggers

Record forecast error deliberately and publish it. Most organisations discard last year's plan when they build this year's, which destroys the only evidence available about whether their model works. A workforce plan that cannot be scored against outcomes cannot improve, and after three cycles it becomes indistinguishable from an opinion.

WHAT TO DO WHEN THE ERROR IS LARGE

A forecast error above 10 per cent is information about the model, not a failure of the planner. Decompose it. Was the task inventory wrong, the multiplier wrong, the volume assumption wrong or the supply assumption wrong. Each points at a different correction and a different owner. An undecomposed error teaches nothing and is usually followed by a return to headcount planning.



A plan nobody can rerun is a forecast. A forecast with owners, thresholds and a rerun rule is a plan.

None of this requires new technology to begin. It requires a task inventory, a named owner per assumption and a rerun rule. Organisations that adopt those three things typically find their first cycle uncomfortable, because the capability gap is larger than the headcount gap and the entry-level arithmetic is worse than expected. That discomfort is the value. The alternative is a number that balances and a workforce that does not.

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